

Software Project Lab - 3 Midterm Presentation



Anatomically **Prior-Enhanced X-Ray Network** for Thoracic Disease Classification and Localisation

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Project Overview

APEX-Net is an AI-powered system that analyses chest X-ray images to automatically detect different thoracic diseases and highlight the affected areas.



**Assist Doctors in
Underserved Areas**



**Catch Rare Diseases
Early**



**Make AI Decisions
Transparent**

Three Critical Gaps in Existing AI Systems

1

**Rare Disease
Blindness**

2

No Anatomical Focus

3

No Visual Explanation

APEX-Net: One System, Three Answers

Balanced Learning

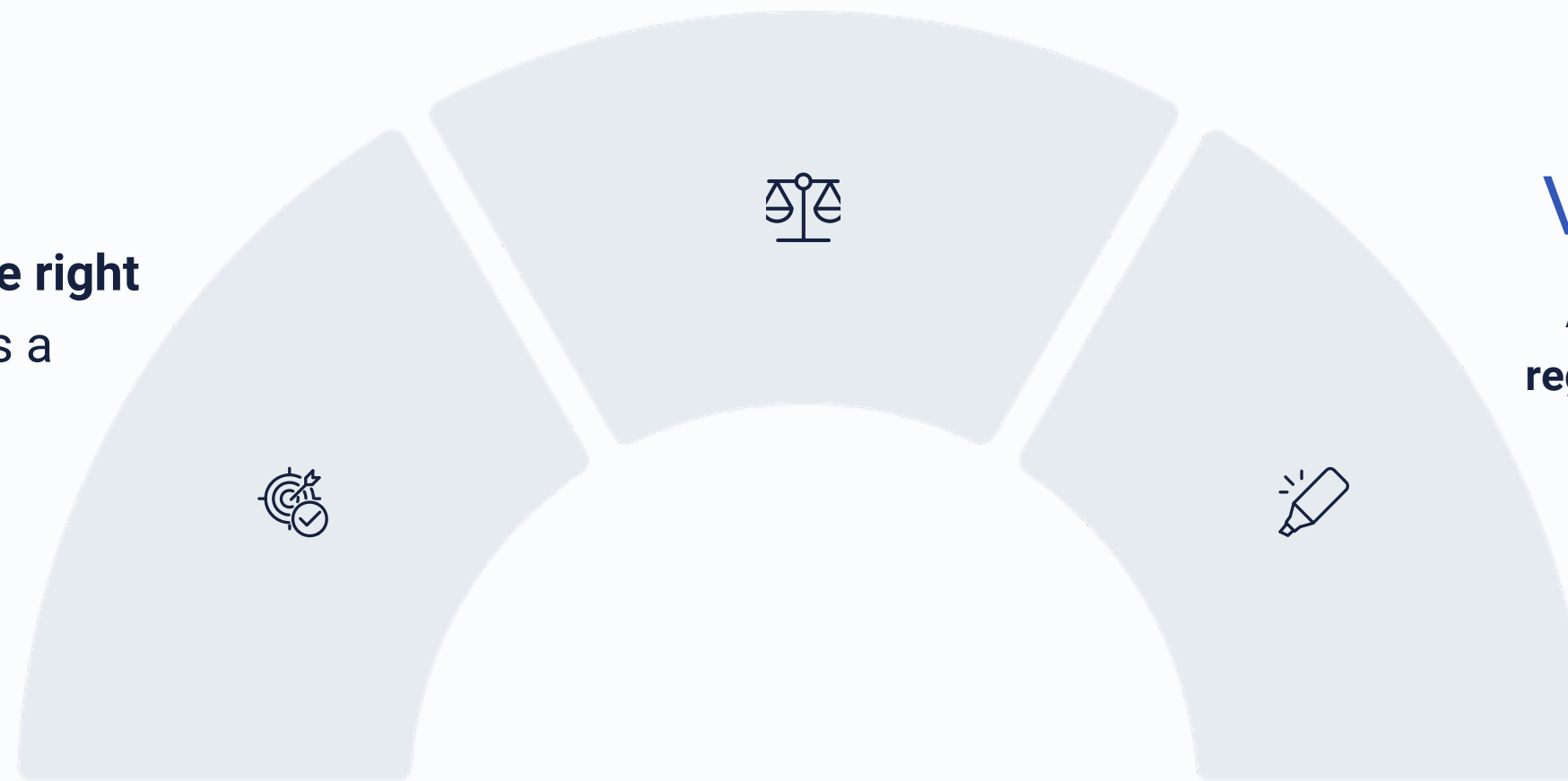
A specialised training strategy that gives **extra emphasis to rare diseases**, reducing missed diagnoses.

Smart Attention

Guides the model to focus on **the right anatomical regions** — just as a radiologist would.

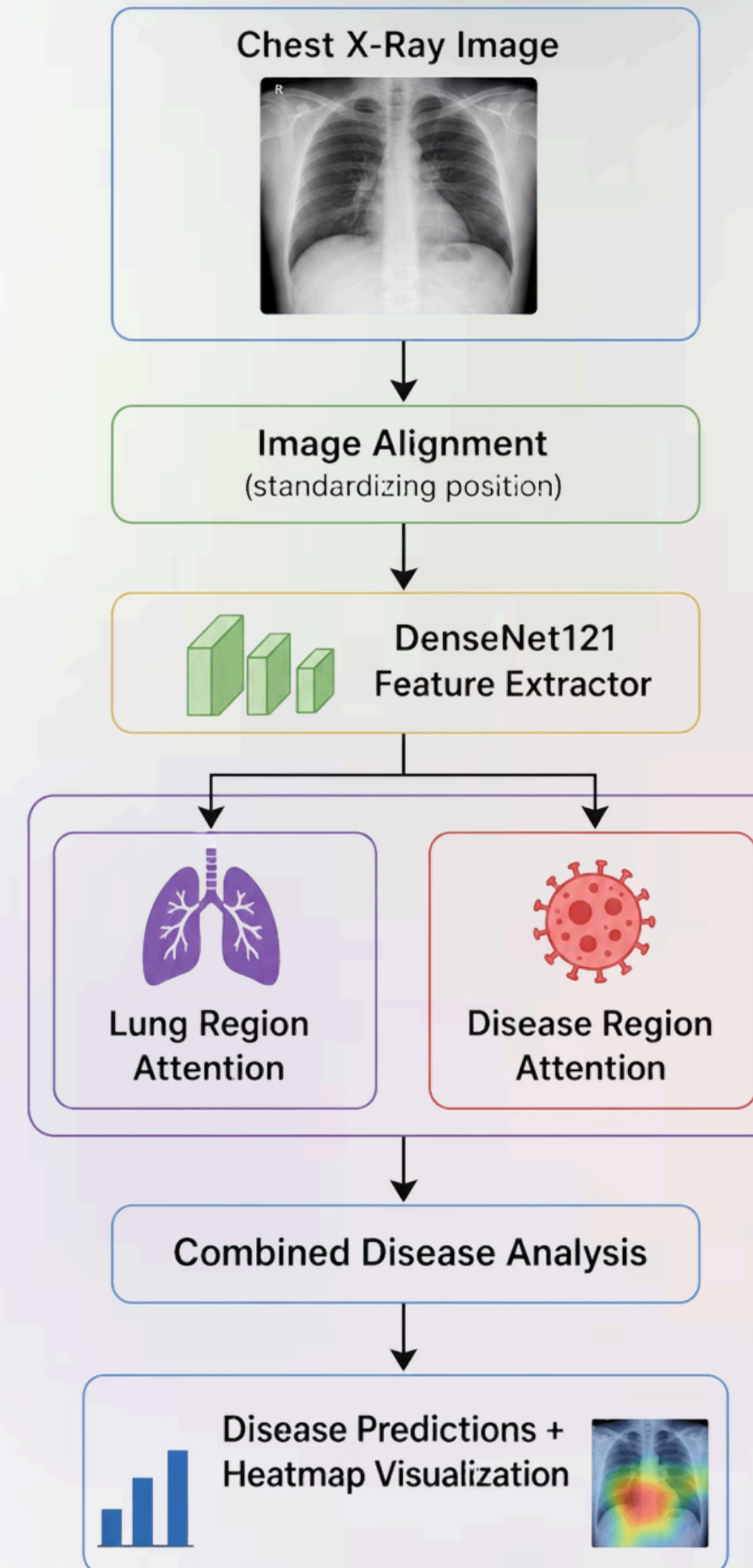
Visual Explanation

Automatically **highlights the affected region** in the X-ray so clinicians can verify every prediction.



METHODOLOGY

How APEX-Net Works



EXPECTED OUTCOMES

What This Project Will Deliver



Disease Detection

Accurately classify **14 distinct chest diseases** from a single X-ray image.



Localisation

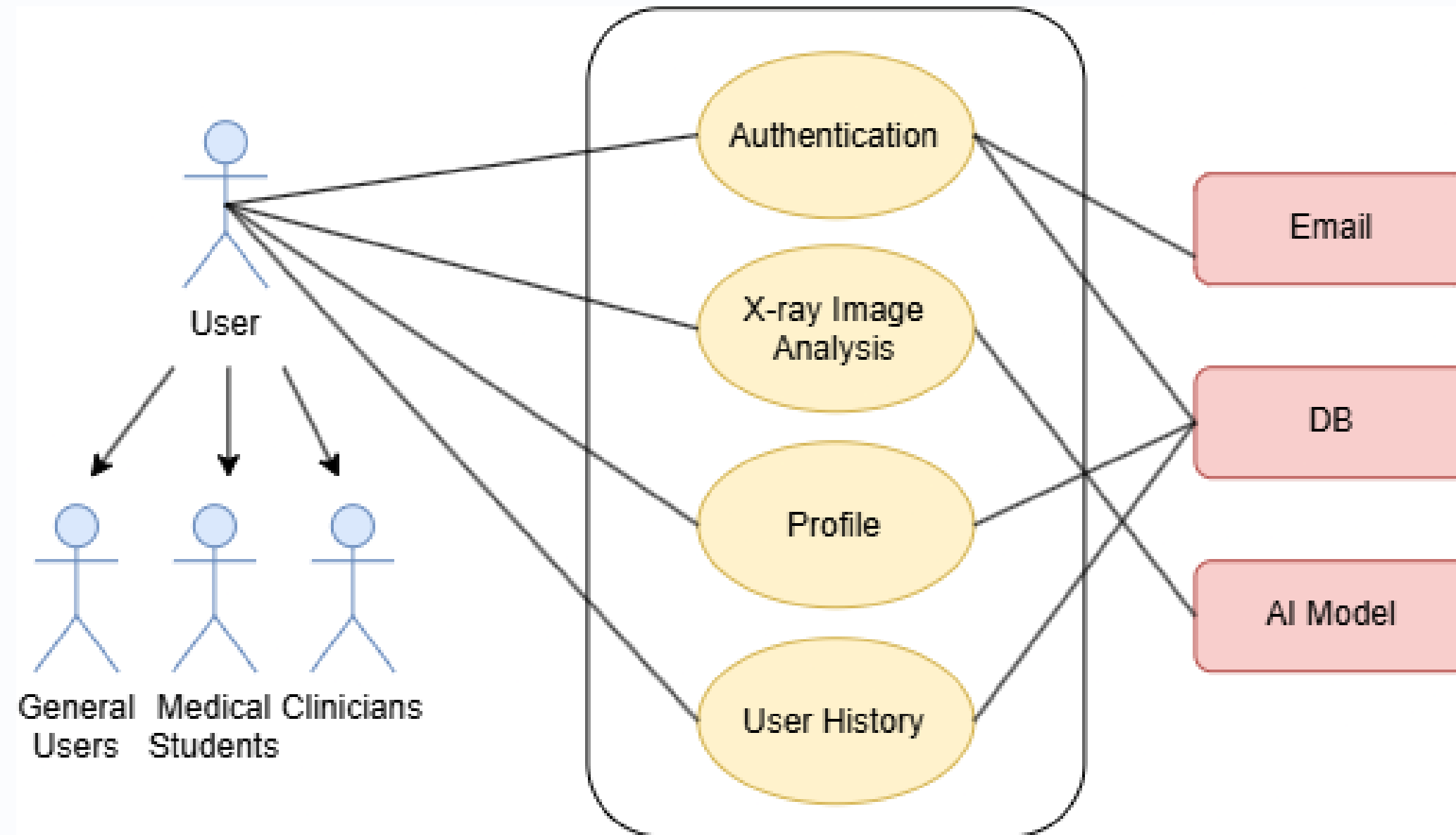
Highlight **exactly where** in the X-ray each detected disease is located.



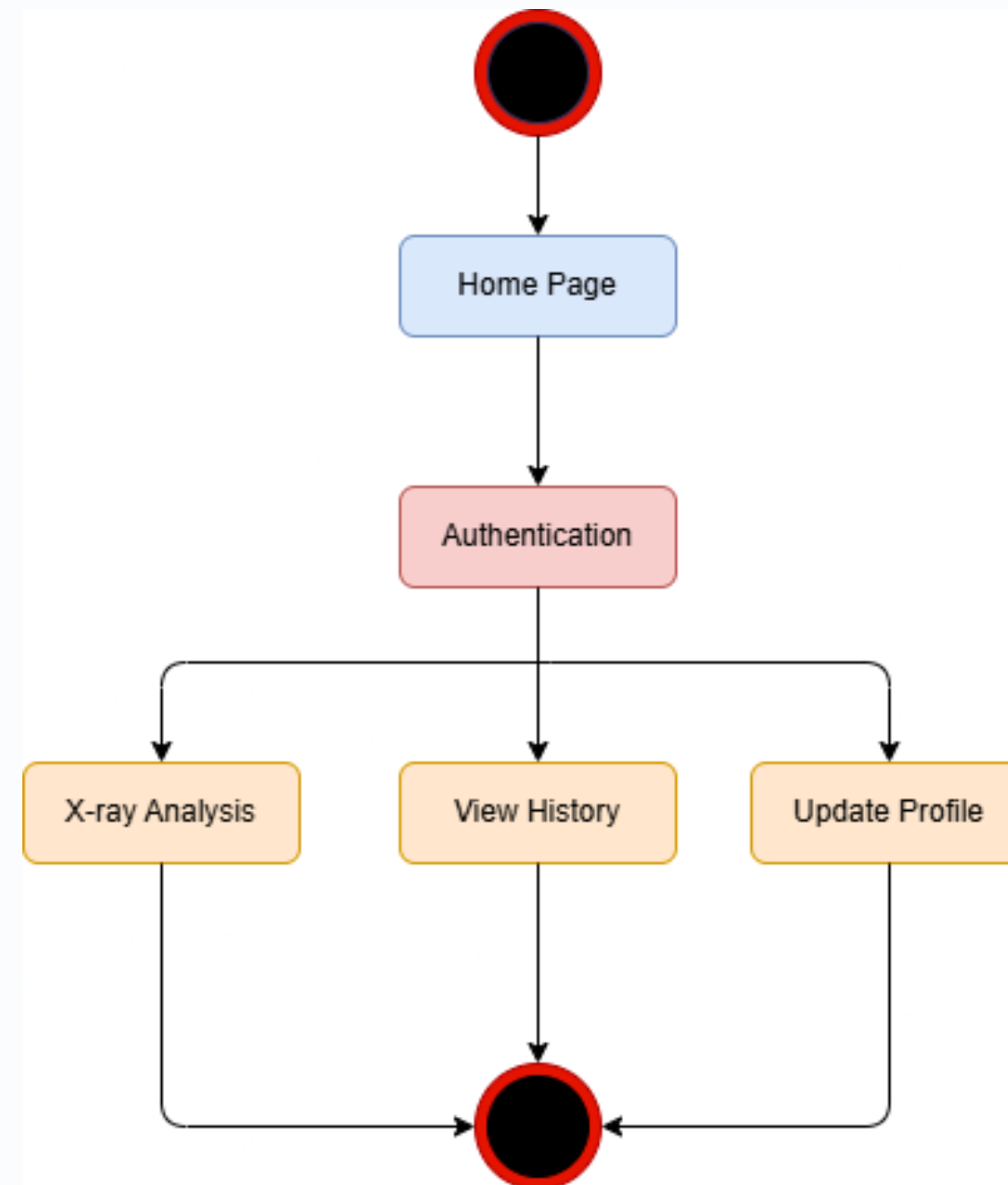
Web Application

A working **demonstration interface** that clinicians or reviewers can interact with directly.

Use Case level 1.0

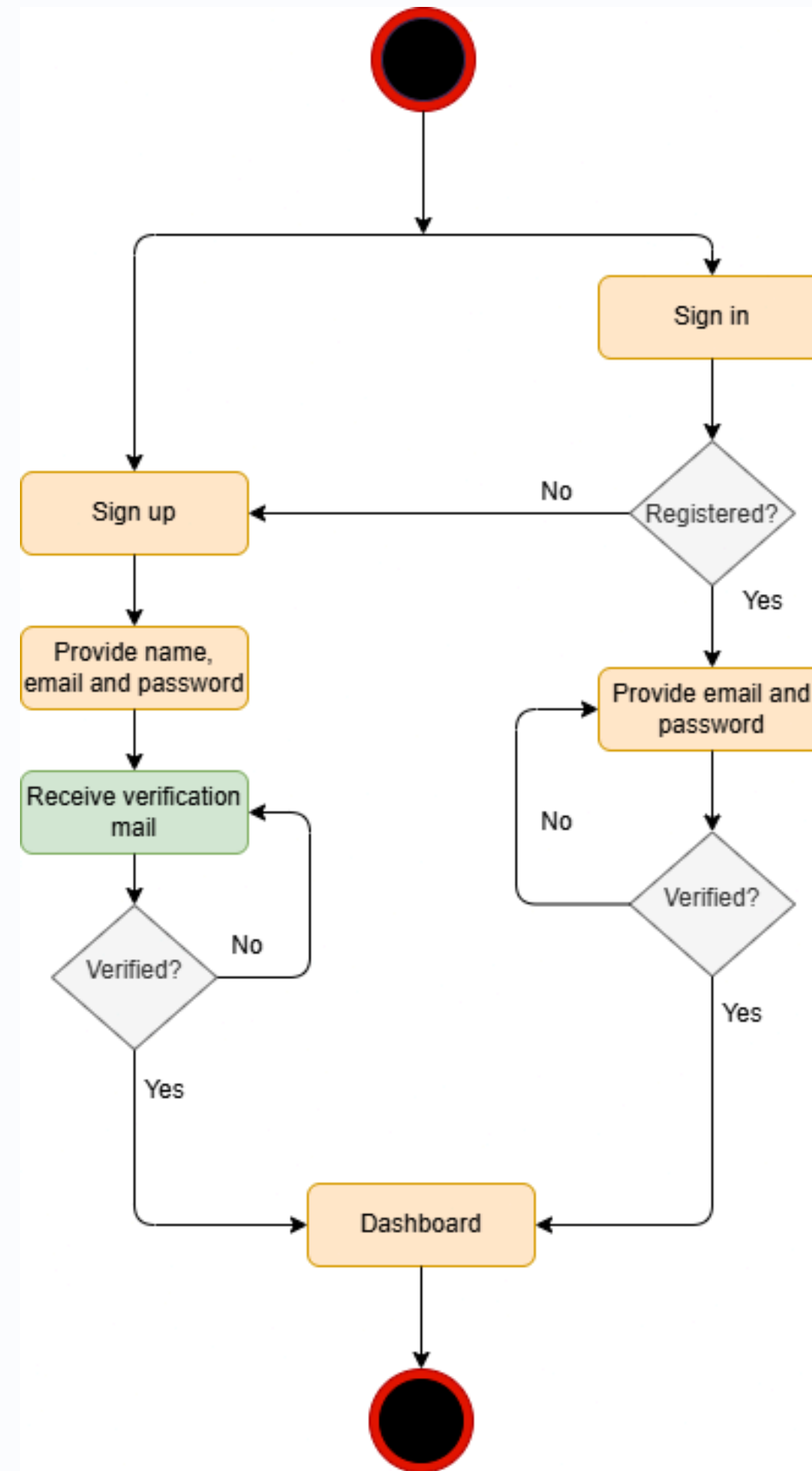


Activity level 1.0



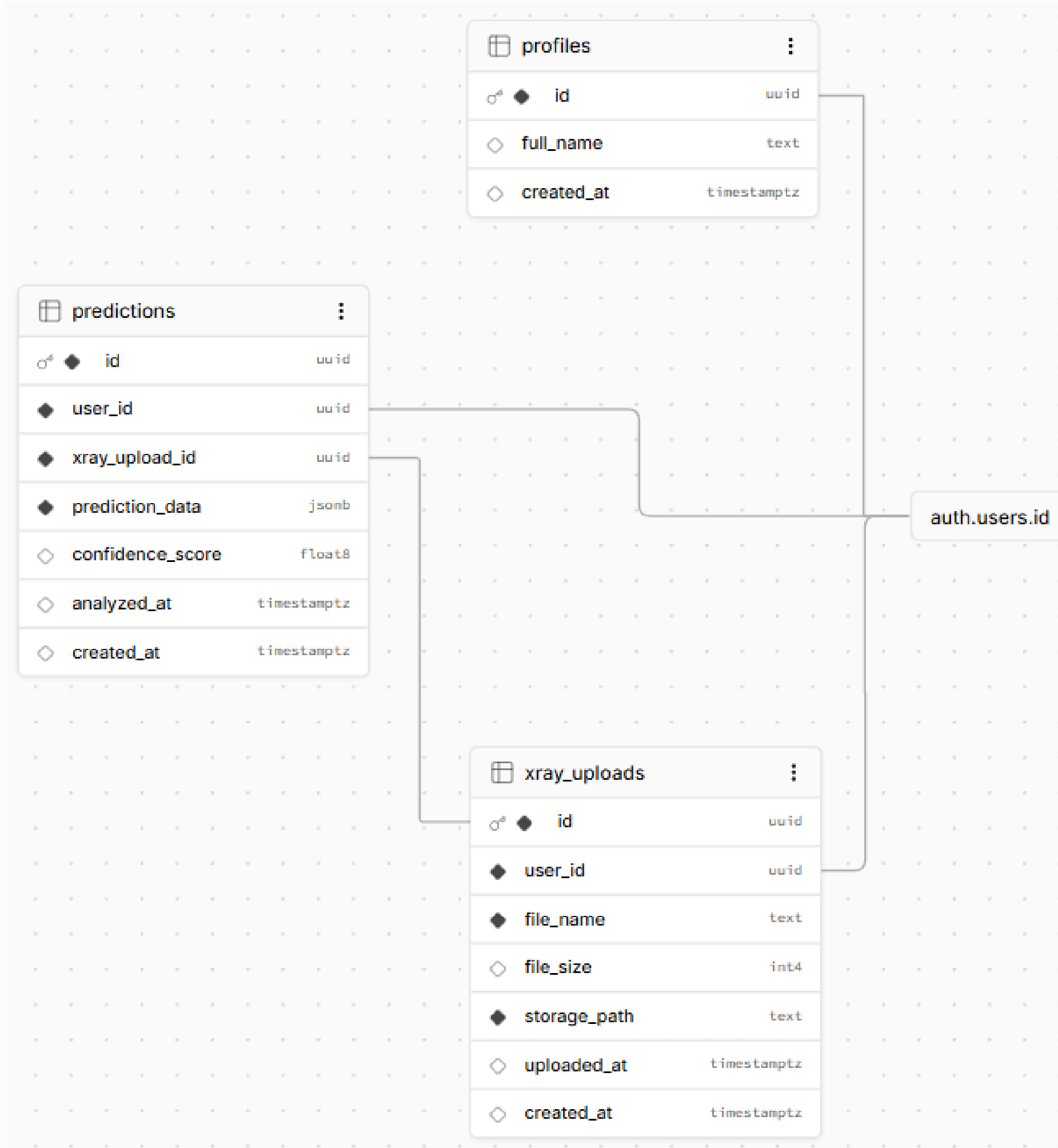
Diagrams

Activity level 1.1 -Authentication



Diagrams

ER









Sign In

Get Started

Clinical-grade AI imaging workflows

See more with
precision-first chest
X-ray analysis.

APEX-Net helps medical teams move from imaging intake to actionable insight with faster review, clearer findings, and confident next-step guidance.

Test X-Ray →

Create account

Realtime insights

Structured findings

Secure workflow

APEX-NET · AI PREVIEW

LIVE



Consolidation · 93%

Opacity · 78%

Left mid-lobe consolidation

Pattern consistent with bacterial pneumonia · 93% confidence



Rapid triage support

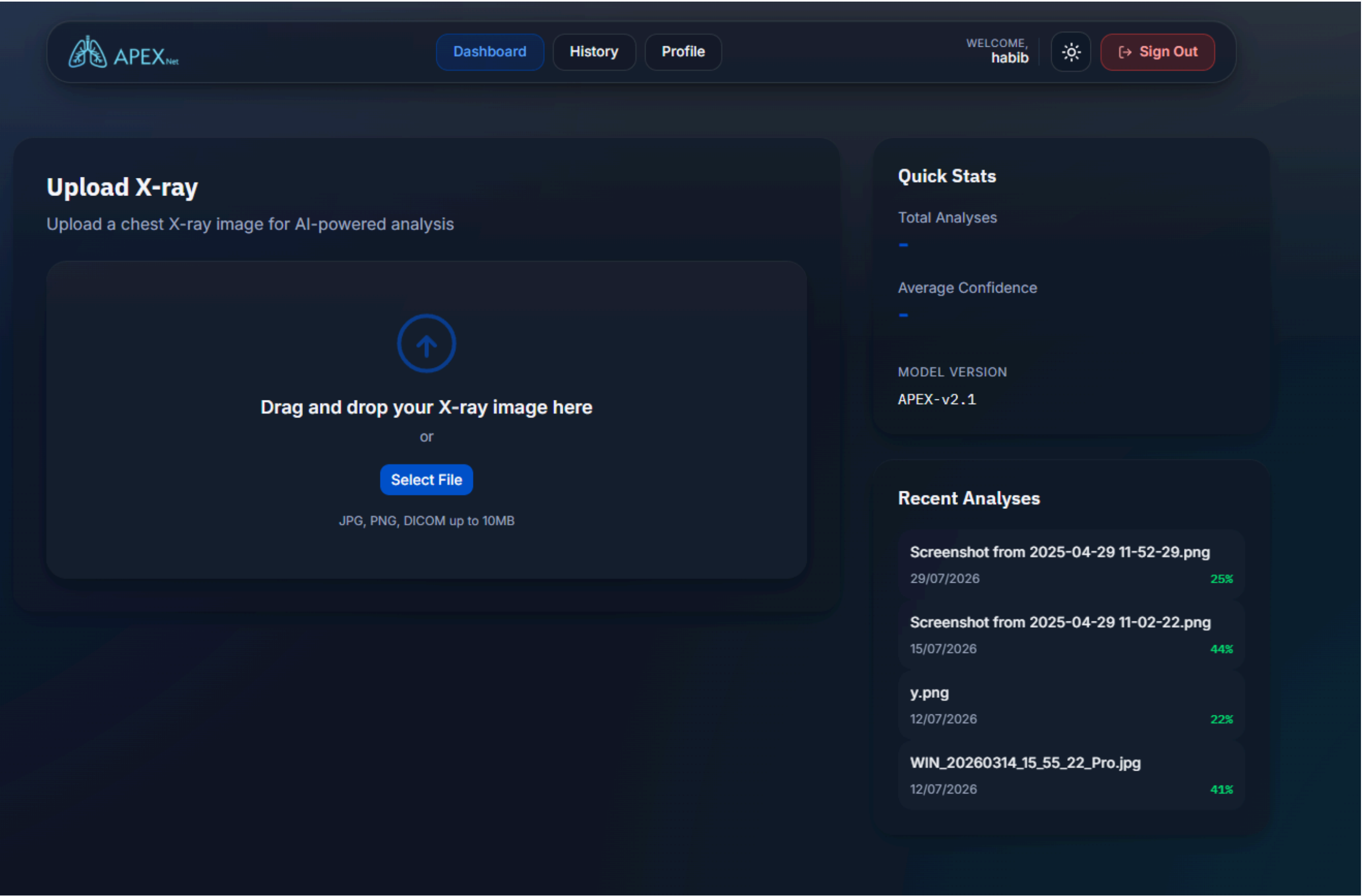


Clinical-grade summaries



Secure collaboration

11



Progress

Model Training

```
❖ (apex_env) PS E:\Projects\APEX-Net\APEX-Net\DacNet> python scripts/dacnet.py --data_dir D:\NIH_data
Using dataset directory: D:\NIH_data
wandb: Loading settings from E:\Projects\APEX-Net\APEX-Net\DacNet\wandb\settings
wandb: Tracking run with wandb version 0.28.1
wandb: W&B syncing is set to `offline` in this directory. Run `wandb online` or set WANDB_MODE=online to enable cloud syncing.
wandb: Run data is saved locally in E:\Projects\APEX-Net\APEX-Net\DacNet\wandb\offline-run-20260801_213255-bspvyjbg
wandb: View this run in the terminal with `wandb leet`
Epoch 1/25 [Train]: 13%|██████████| 1646/12981 [30:53<3:37:01, 1.15s/it, loss=0.0523]
```

Thank You